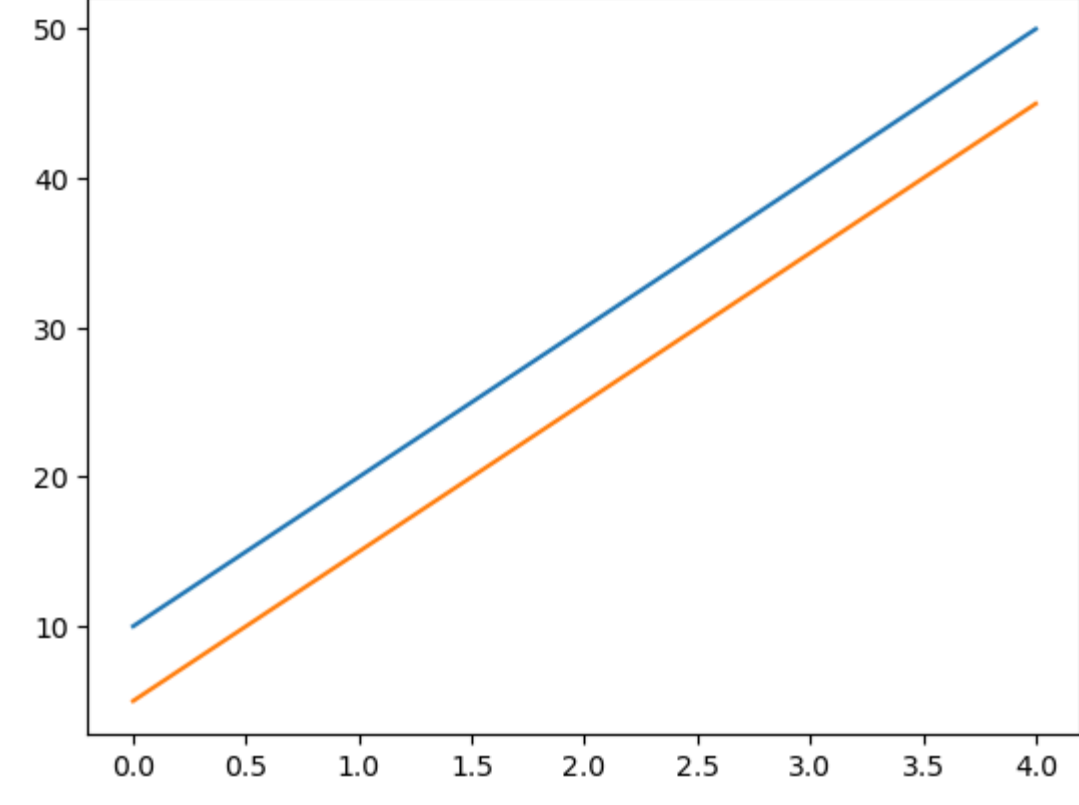
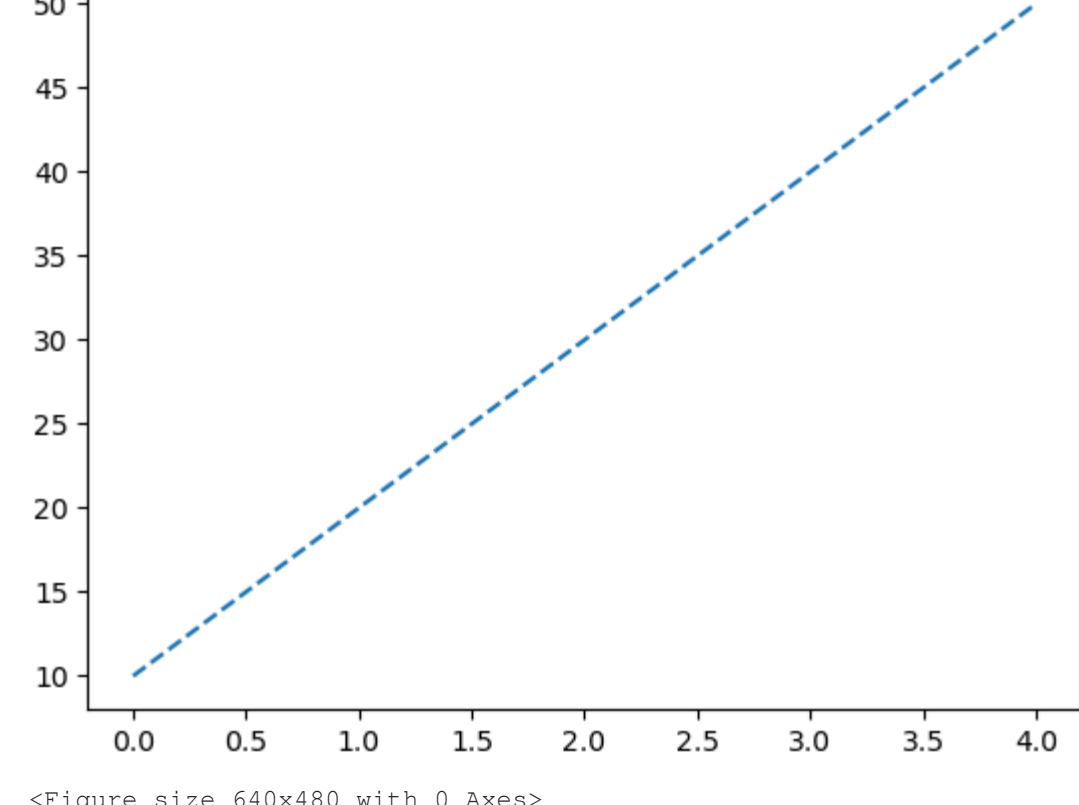


```
MATPLOTLIP
In [1]: import matplotlib as mlp
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
```

```
In [3]: x = [ 10, 20, 30, 40, 50 ]
y = [ 10, 15, 20, 30, 40 ]
plt.plot(x)
plt.plot(y)
plt.show()
```



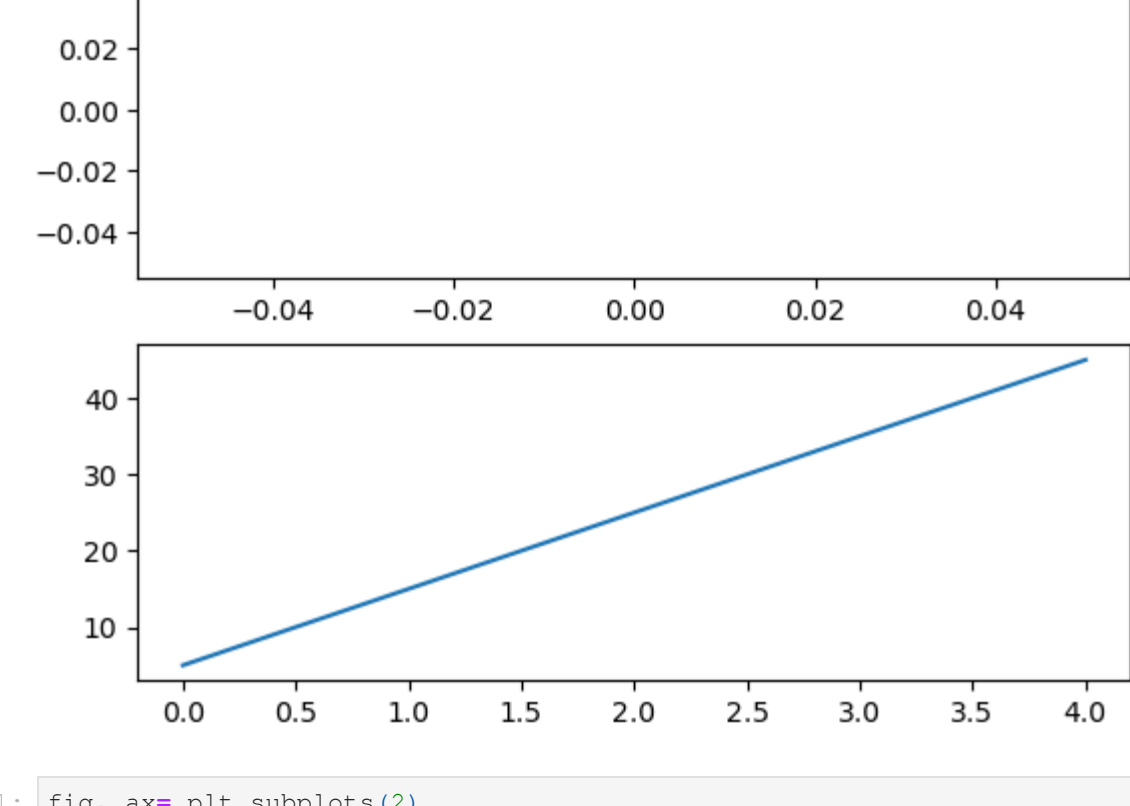
```
In [5]: plt.plot(x,'--')
fig = plt.figure()
```



<Figure size 640x480 with 0 Axes>

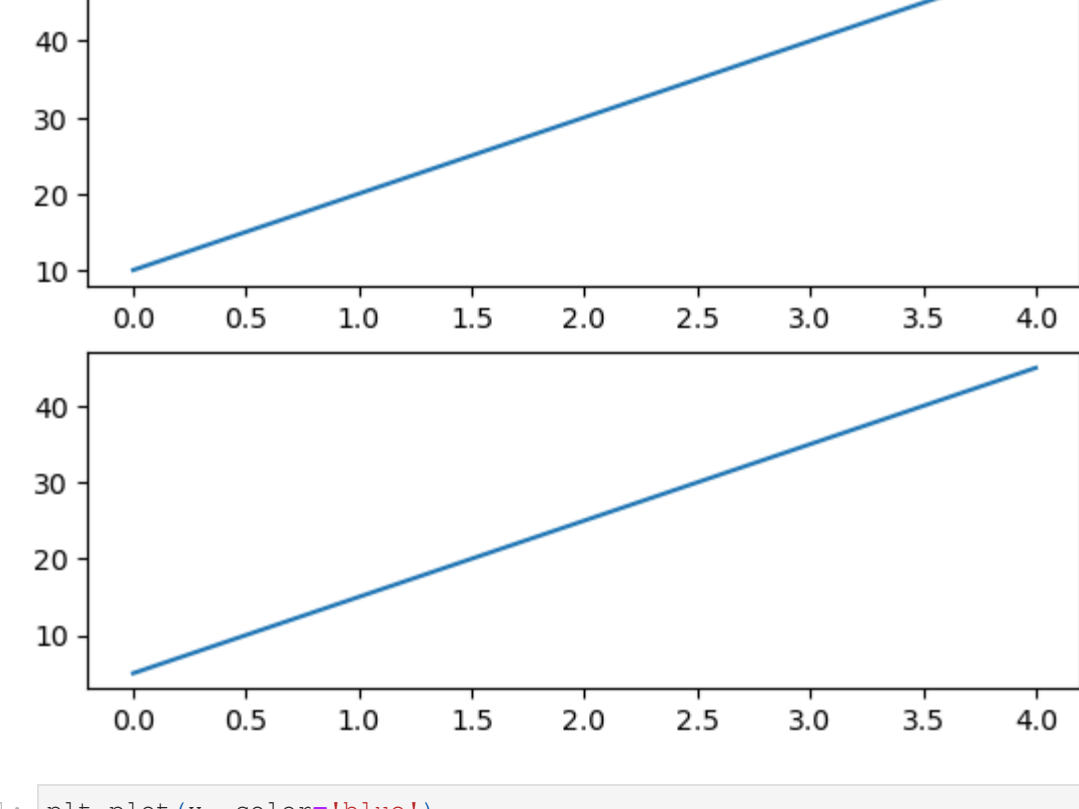
```
In [6]: plt.figure()
plt.subplot(2, 1, 1)
plt.plot()
plt.subplot(2,1,2)
plt.plot(y)
```

Out[6]: <matplotlib.lines.Line2D at 0x1b5d54fe4e50>



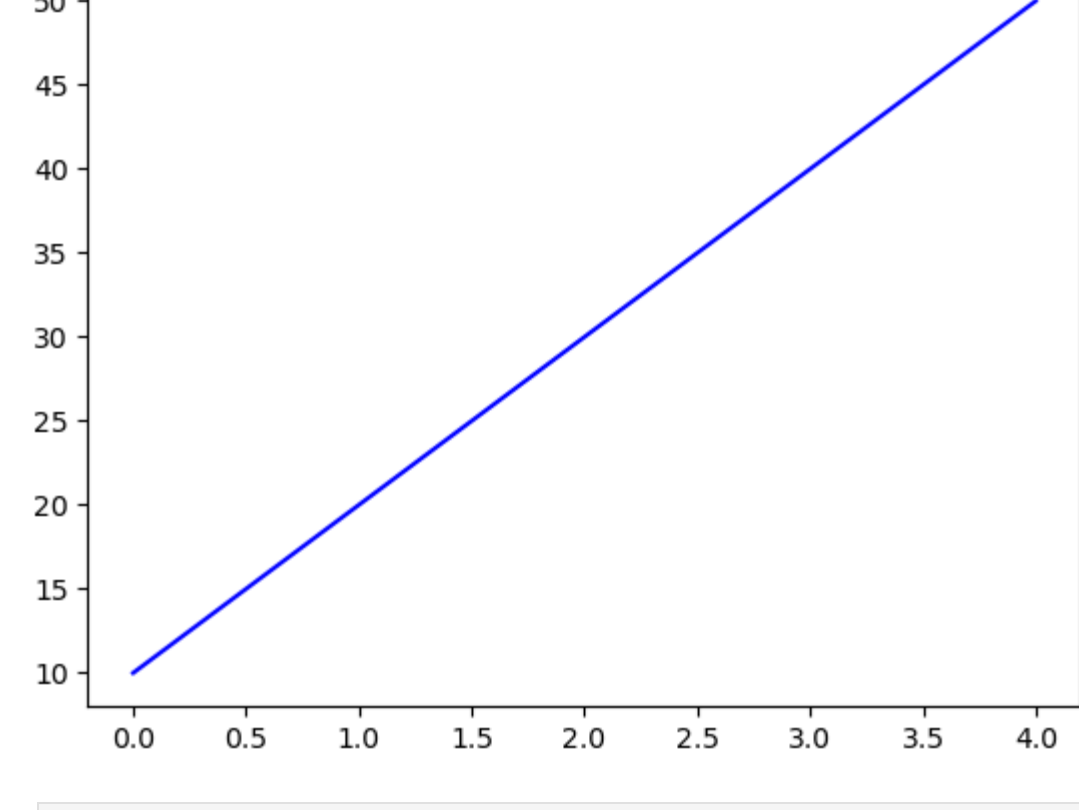
```
In [8]: fig, ax= plt.subplots(2)
ax[0].plot(x)
ax[1].plot(y)
```

Out[8]: <matplotlib.lines.Line2D at 0x1b5d57ed310>



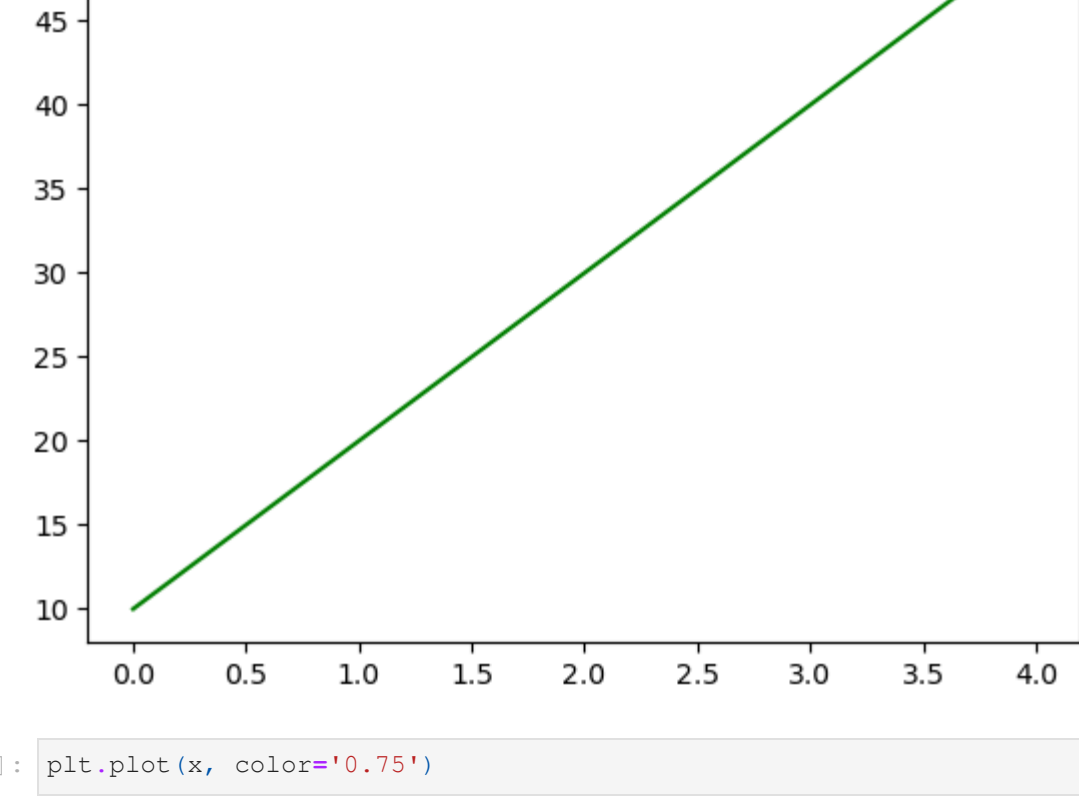
```
In [9]: plt.plot(x, color='blue')
```

Out[9]: <matplotlib.lines.Line2D at 0x1b5d54cd7110>



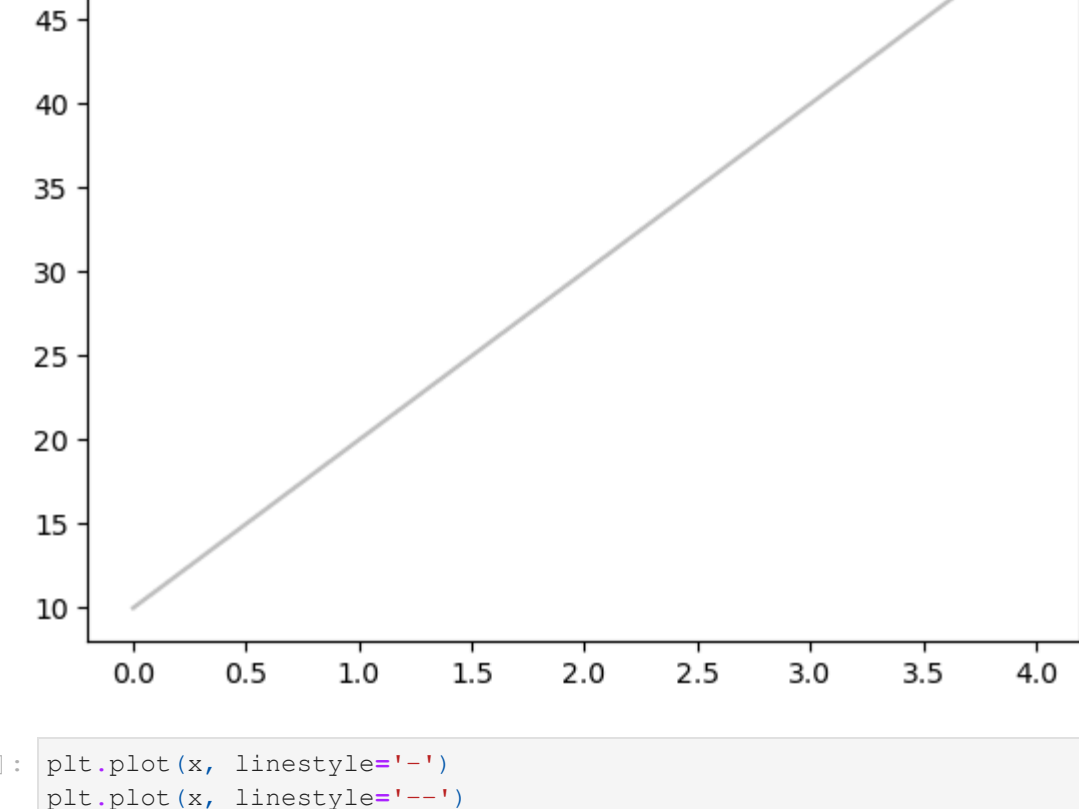
```
In [11]: plt.plot(x, color='g')
```

Out[11]: <matplotlib.lines.Line2D at 0x1b5d568e8b90>



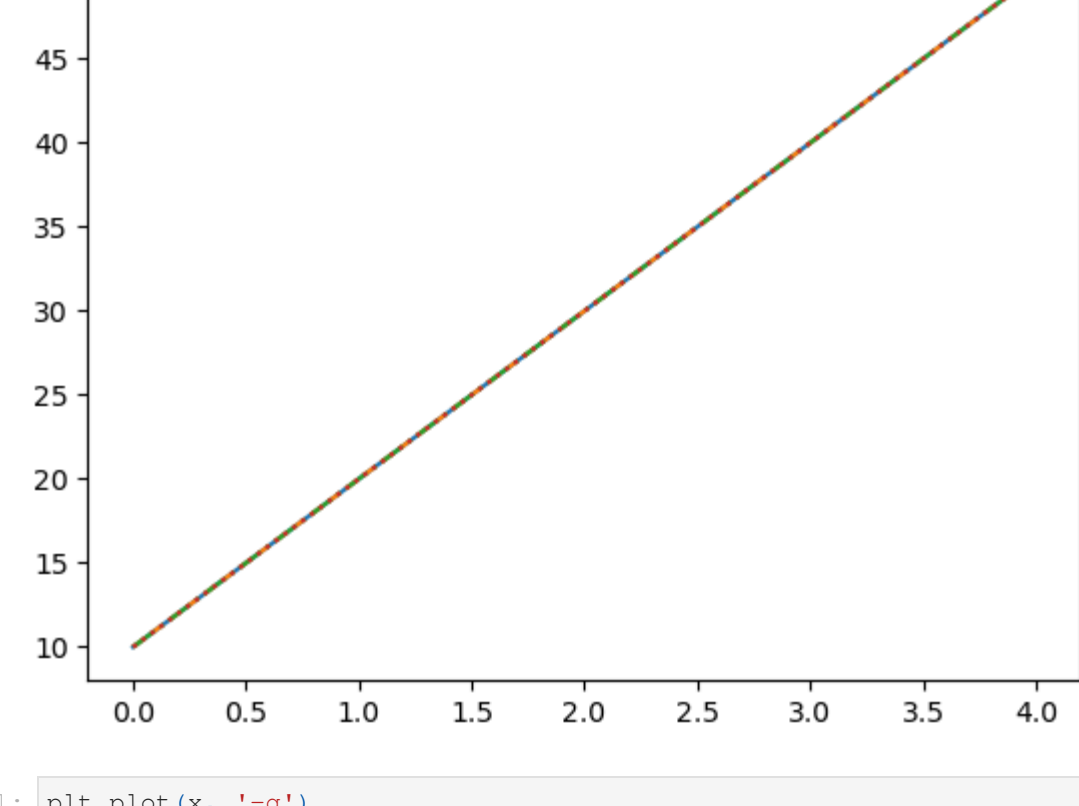
```
In [10]: plt.plot(x, color='0.75')
```

Out[10]: <matplotlib.lines.Line2D at 0x1b5d56f0050>



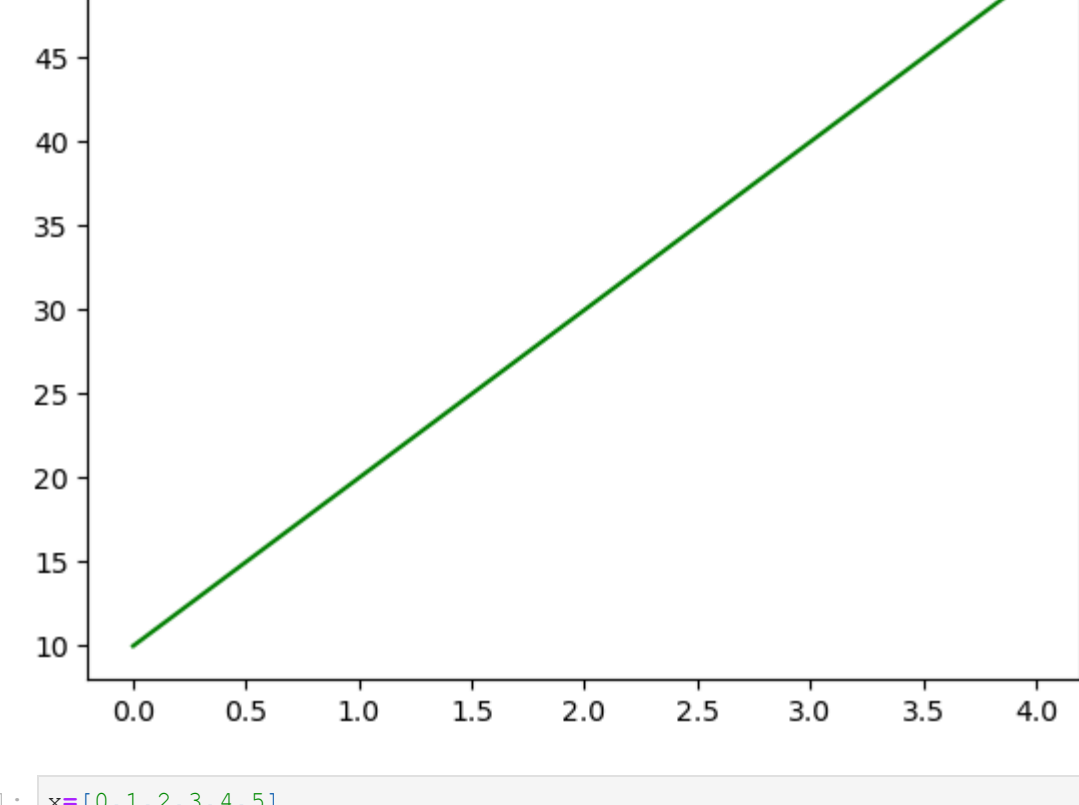
```
In [13]: plt.plot(x, linestyle='-')
plt.plot(x, linestyle='--')
plt.plot(x, linestyle=':')
plt.plot(x, linestyle=':')
plt.plot(x, linestyle=':')
plt.plot(x, linestyle=':')
```

Out[13]: <matplotlib.lines.Line2D at 0x1b5d56bfdd10>

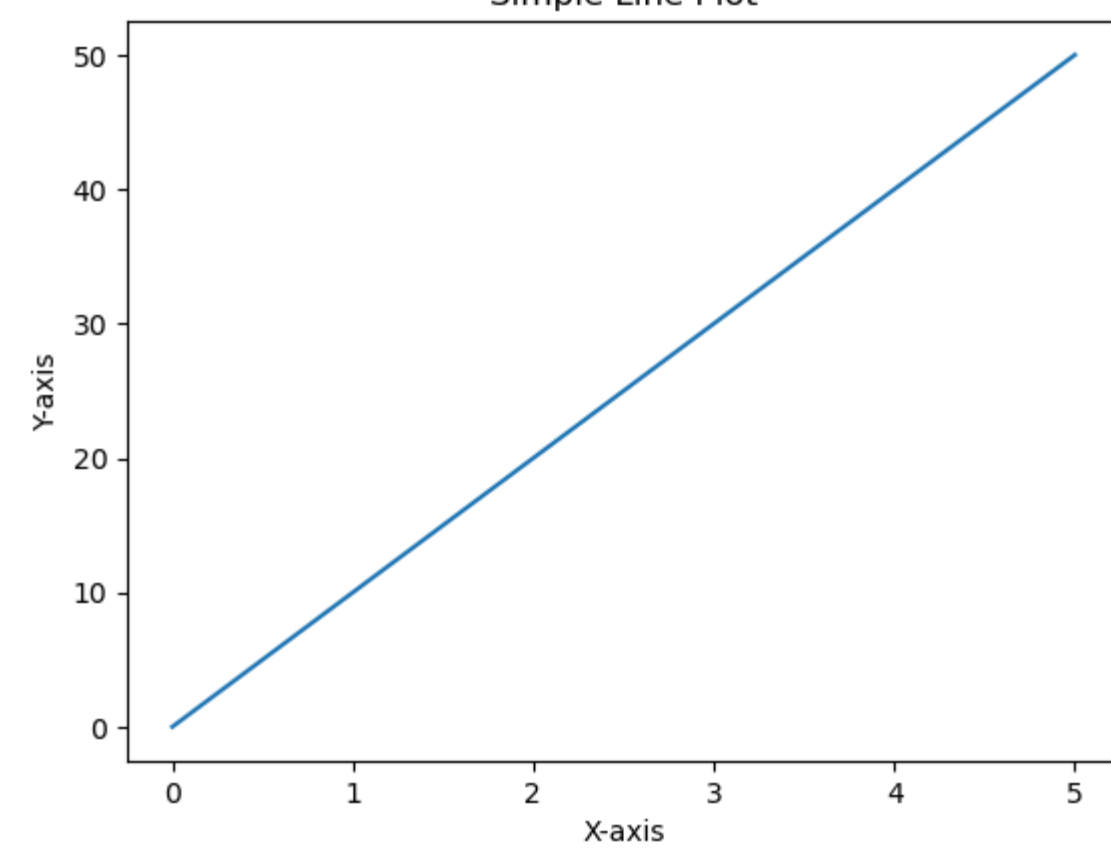


```
In [14]: plt.plot(x, 'g')
```

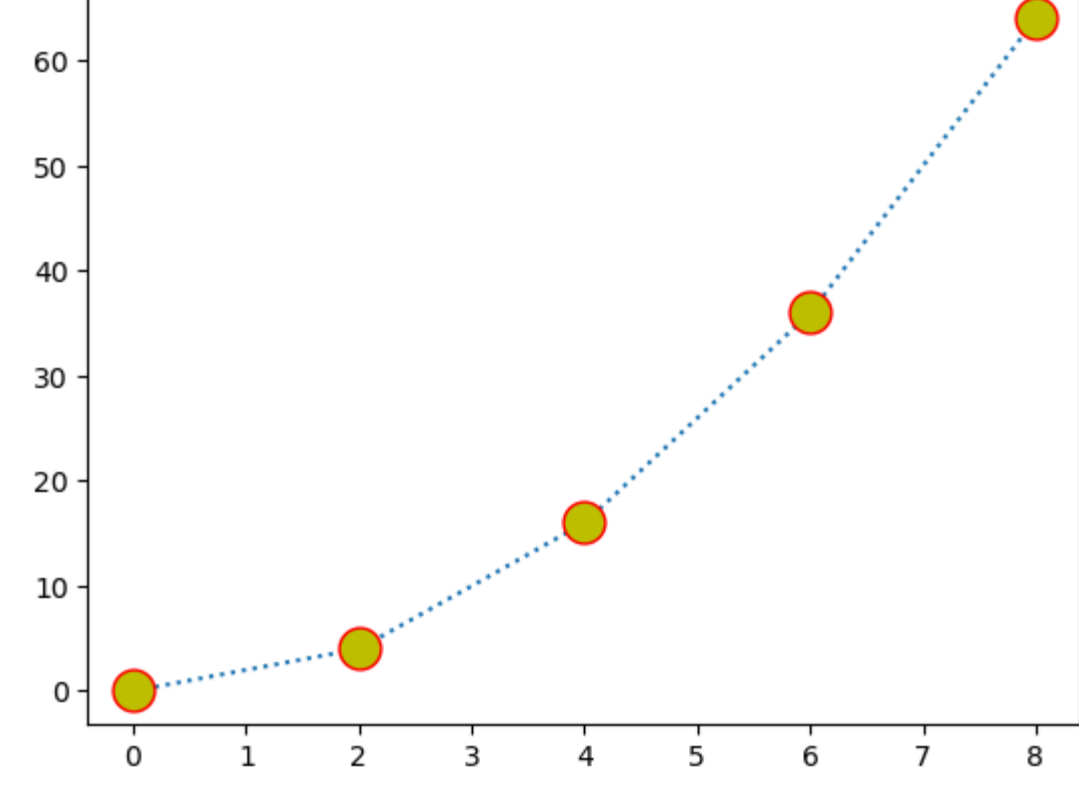
Out[14]: <matplotlib.lines.Line2D at 0x1b5d56dc2850>



```
In [16]: x=[0,1,2,3,4,5]
y=[0,10,20,30,40,50]
plt.plot(x,y)
plt.title("Simple Line Plot")
plt.xlabel("X-axis")
plt.ylabel("Y-axis")
plt.show()
```

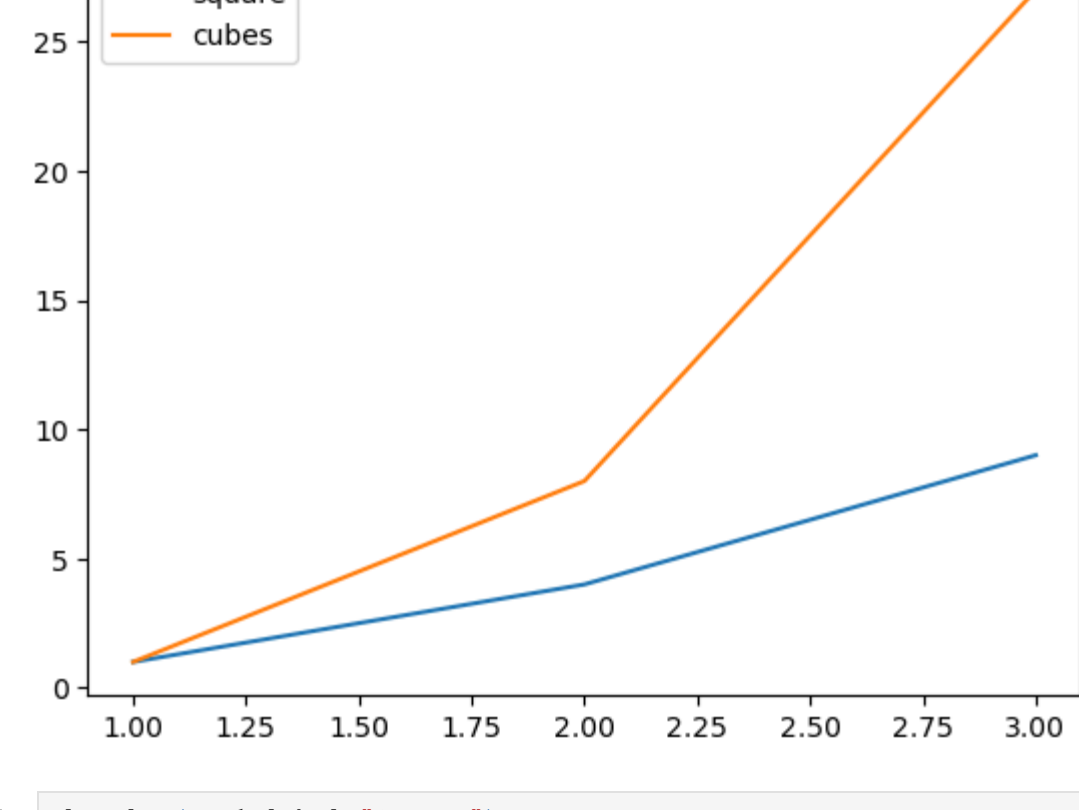


```
In [22]: x=np.arange(0,10,2)
y=x**2
plt.plot(x,y, marker="o", ms=15, mec='r', mfc='y', linestyle='dotted')
plt.show()
```

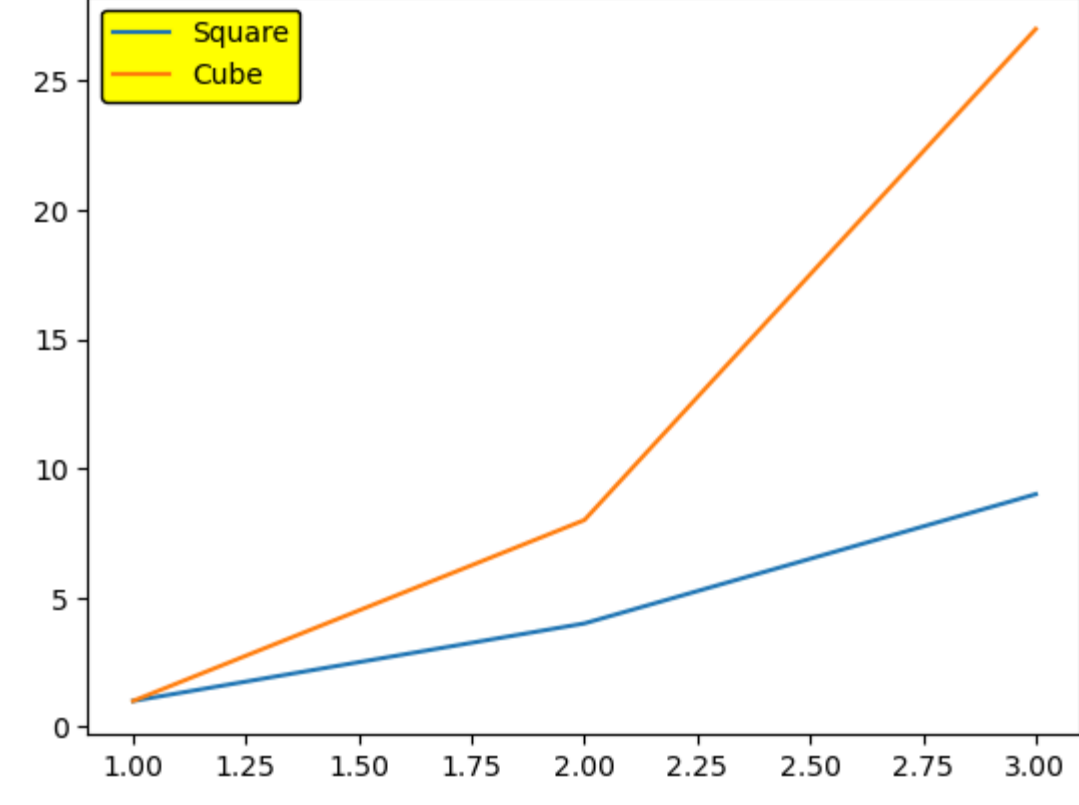


```
In [25]: x=[1,2,3]
y1=[1,4,9]
y2=[1,8,27]
plt.plot(x,y1,label="square")
plt.plot(x,y2,label="cubes")
plt.legend()
plt.show()
```

Out[25]: <matplotlib.legend.Legend at 0x1b5d4c712b0>

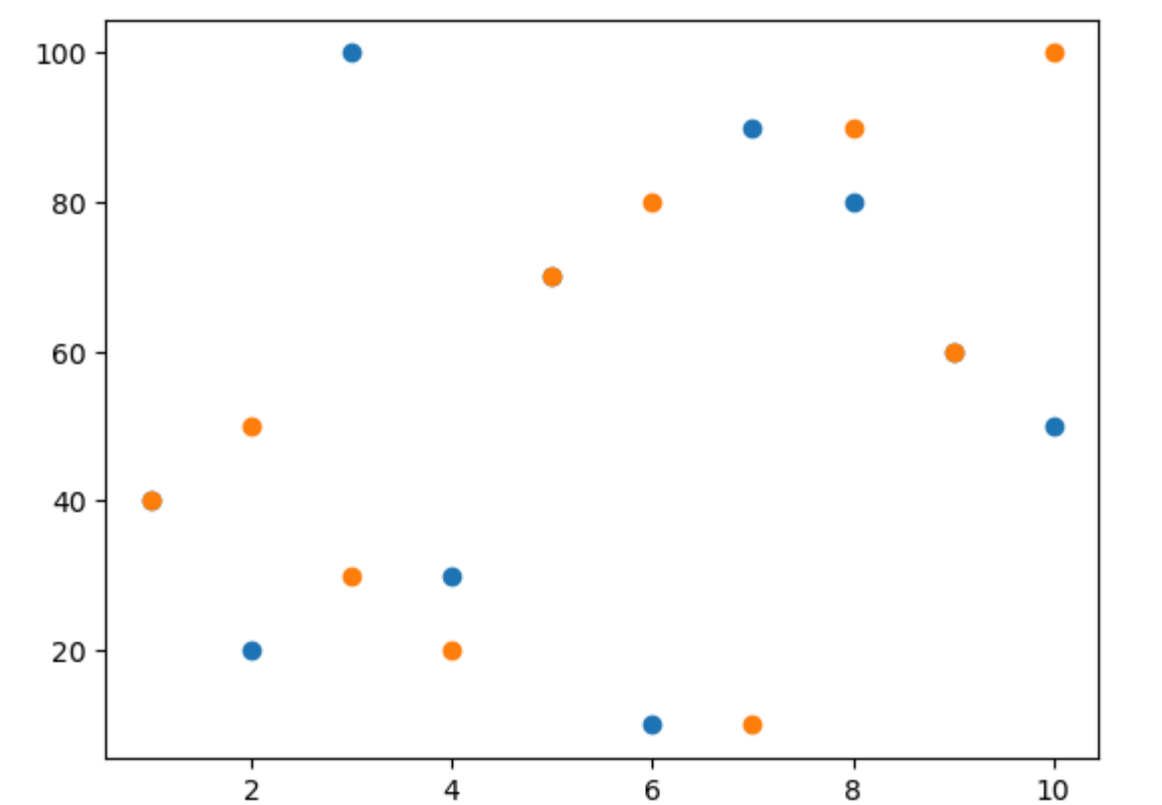


```
In [26]: plt.plot(x,y1,label="square")
plt.plot(x,y2,label="cubes")
plt.legend(loc='upper left', framealpha=1, facecolor="yellow", edgecolor="black")
plt.show()
```

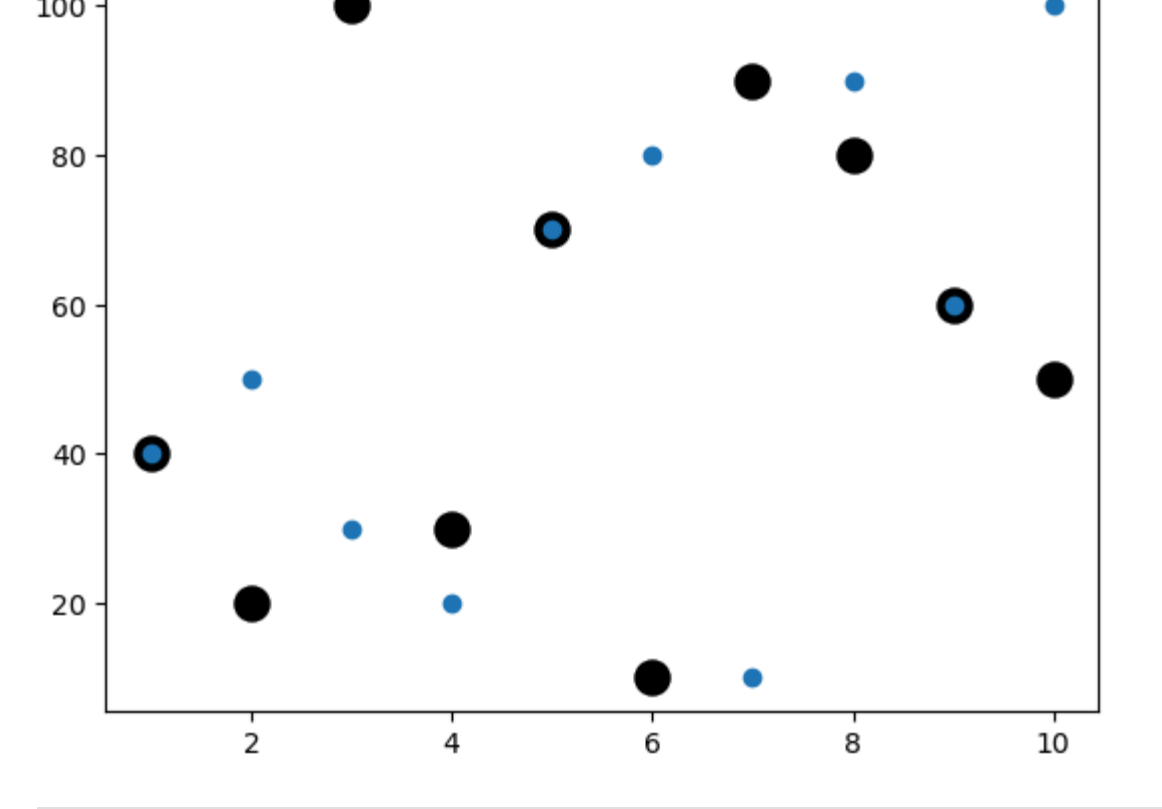


```
In [27]: import random
x=[1,2,3,4,5,6,7,8,9,10]
y=[10,20,30,40,50,60,70,80,90,100]
random.shuffle(y)
print(x,y)
plt.scatter(x,y)

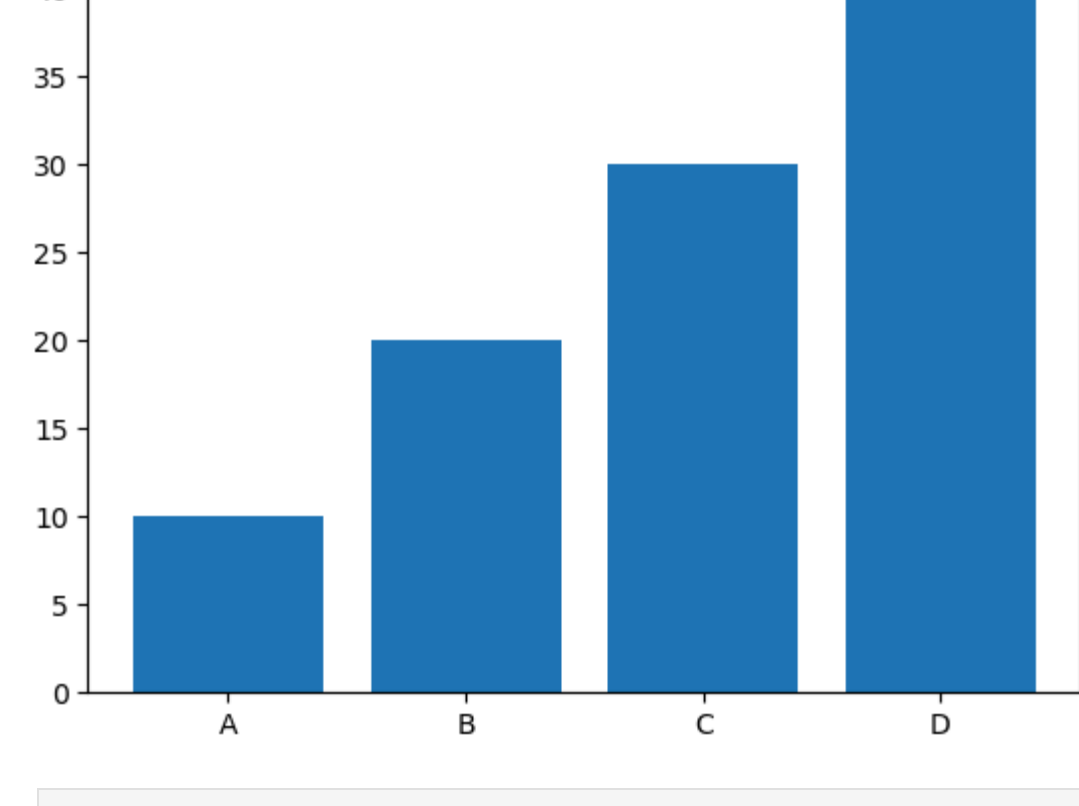
x=[1,2,3,4,5,6,7,8,9,10]
y=[10,20,30,40,50,60,70,80,90,100]
random.shuffle(y)
print(x,y)
plt.scatter(x,y)
plt.show()
```



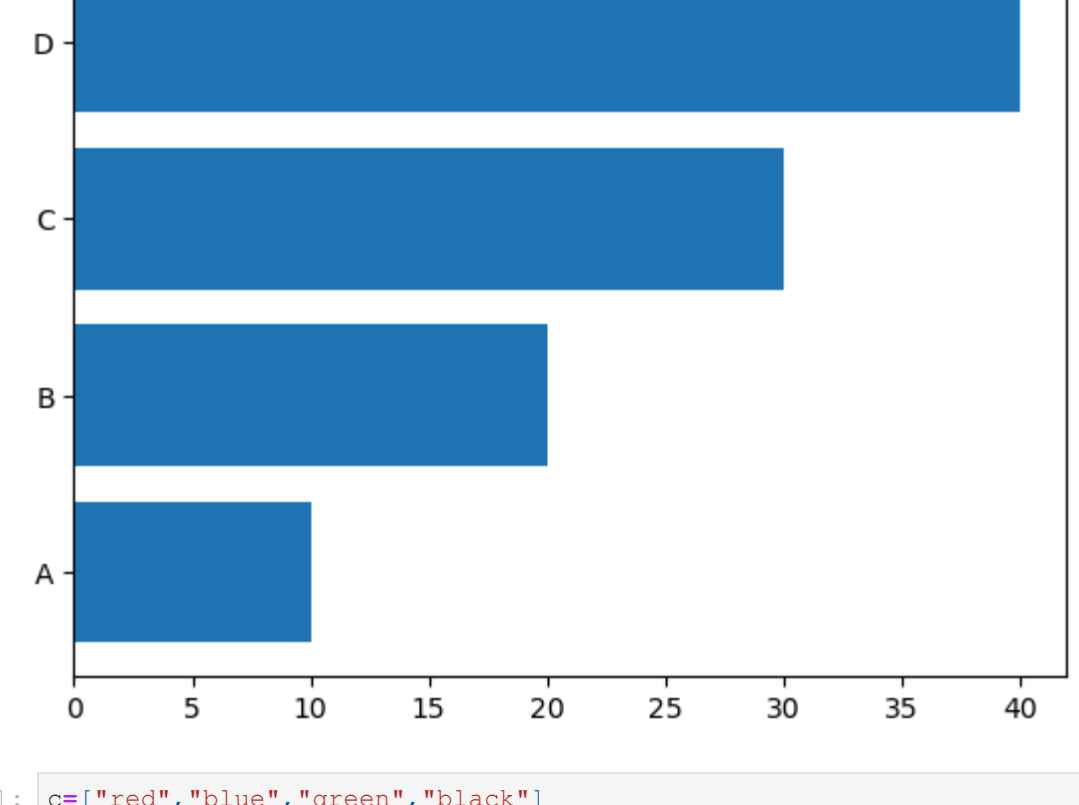
```
In [29]: print(x,y)
plt.scatter(x,y,color="black", marker="o",s=150)
plt.scatter(x1,y1)
plt.show()
```



```
In [29]: x=['A','B','C','D']
y=[10,20,30,40]
plt.bar(x,y)
plt.show()
```



```
In [30]: x=['A','B','C','D']
y=[10,20,30,40]
plt.barh(x,y)
plt.show()
```



```
In [31]: cm="red","blue","green","black"
plt.bar(x,y,color=cm,width=0.2)
```

Out[31]: <matplotlib.container object of 4 artists>

